



EDUCATION				
Degree	University	Specialization	%/CGPA	Year
Master of Science (M.S)	Indian Institute of Technology, Madras	Robotics and Automation	8.19 CGPA	2025 - present
Bachelor of Technology (B. Tech)	Vinoba Bhave University, Hazaribagh	Mechanical Engineering	78.7 %	2018 - 22
Achievements	● Won Best Paper Award at IEEE SPACE 2026 Conference			
PROFESSIONAL EXPERIENCE (3 YEARS 7 MONTHS)				
Project Associate, (ICSR, IITM) Jan'25 - Present	Vehicle On/Off-Road Classification, Free-Space Mapping Feature Development for ADAS Perception Pipeline - Involved in the development of rule-based algorithm to assign a label to the indicate whether the traffic vehicle lies within or outside the road boundary (Execution time: < 100ms, Accuracy: >97%). - Focused on efficient road-lane mappings and static road features to improve classification accuracy. - Free-space boundary mapping around the ego vehicle using its surrounding traffic participants. - Awarded <u>two high customer satisfaction</u> certificates from leading Japanese automotive client.			
Project Associate, (ICSR, IITM) July'24 - Jan'25	Modeling and Performance Analysis of a Closed-Loop Hydraulic System for High-Speed Take-off and Landing of Aircraft (DRDO Project) - Conducted performance analysis of a complete closed-loop hydraulic system for Unmanned Aerial Vehicle (UAV). - Modelled and simulated the hydraulic actuation circuit governing landing gear retraction and extension under high-speed take-off, landing and cruise conditions. - Simulated the system against real flight-condition data and validated its performance to establish operational reliability. - Designed and tuned a PID control strategy for hydraulic valve actuation to achieve stable and responsive system behaviour. - Worked on Simcenter Amesim, PID Controller, Hydraulic System, Reliability Analysis.			
Project Associate, (CSIR, CMERI) June'23 - July'24	Design and Development of an Autonomous Underwater Vehicle (AUV) and its Underwater Docking Station (NIOT Project) - Designed the AUV hull and optimized its geometry for minimum hydrodynamic drag through Computational Fluid Dynamics (CFD). - AUV docking station with a funnel-type homing structure and latching mechanism for autonomous capture and underwater charging. - Integrated USBL, DVL and INS units using sensor fusion to enable reliable underwater navigation and communication. - Developed and tested a Remotely operated vehicle (ROV) at the institute's shallow basin facility, including a magnetic wheel system for ship hull cleaning and a non-destructive Testing (NDT) sensor mechanism for hull inspection. - Worked on Ansys Fluent, Ultra-Short Baseline (USBL), Dopler Velocity Log (DVL), Inertial Navigation System (INS).			
Graduate Apprentice Trainee, (CSIR, NML) Jan'23 - Jun'23	Process Optimization and Post-Process Characterization of Wire Arc Additive Manufacturing (WAAM) for Metallic Components. - Conducted Research and development on Wire Arc Additive Manufacturing (WAAM) processes for the fabrication of metallic components. - Designed and executed experiments to optimize WAAM process parameters across multiple metal alloys. - Performed post-process characterization, including residual stress measurement by the contour method and dimensional accuracy assessment using a Coordinate Measuring Machine (CMM). - Collaborated with academic and industrial partners to apply WAAM to real-world engineering applications. - Worked on Additive Manufacturing, Material Characterization, Coordinate Measuring Machine (CMM).			
Intern (BCCL, Coal India) Mar' 21- April' 21	- Worked on the assembly of gearboxes and internal combustion engine units as per standard operating procedures, gaining hands-on experience in mechanical assembly and shop-floor practices. - Performed component testing and supported quality assurance activities to verify reliability before integration into larger vehicle systems.			

RESEARCH EXPERIENCE			
Supervisors: Dr. Asokan T (Autonomous Systems Lab)			
Research Scholar, IIT Madras Feb'24 - present	Development of Perception Capabilities for Autonomous Driving Applications (Project Page)		
	- Performed sensor integration and registration; collected time synchronized multimodal data. - Developed Detect-Classify-Track pipeline for the ADAS applications using camera, LiDAR, IMU, GPS. - Achieved 62% classification accuracy with execution time < 0.5 seconds for the perception pipeline. - Implemented vehicle localization on OpenStreetMap using GPS and IMU data.		
	Forward Prediction of Possible Trajectories of On-Road Traffic Vehicles (Project Page)		
	- Proposed MC-EKF³ framework for trajectory prediction, evaluated using Ackermann steering models. - Experimented using multiple motion models, achieved minimum average displacement error of 0.101 m. - Accepted at upcoming IEEE International conference on Smart Technologies for Power, Energy and Control, 2025.		
	Fusion of Geometric and Kinematic Features for LiDAR-Based Vehicle Classification (Project Page)		
	- Trained machine learning models with feature augmentation to improve classification accuracy. - Improved classification accuracy by 5% on ambiguous classes, with 89% overall accuracy on NuScenes dataset. - Ongoing work: extending to deep neural networks for trajectory prediction conditioned on vehicle class.		
COURSE PROJECTS			
Modern Computer Vision (EE5178)	Detection and Classification of Mosquito Species in Low Light Environment (Team Ranking: 9/23; mAP: 0.59472) Used ZERO-DCE neural network for low light enhancement and YOLO for detection and classification of species.		
Deep Learning for Imaging (EE5179)	Image Restoration from Noisy Images while Preserving Defects (Team Ranking: Top 3; SSIM: 0.929, PSNR: 29.35dB) Evaluated multiple deep learning architectures for image restoration.		
	Binary Adder Implementation using Recurrent Neural Networks		
	Curated dataset and implemented LSTM, GRU models to learn and perform addition operation.		
COURSE WORK AND SKILLS			
- Introduction to machine learning (EE5180) - Deep learning for imaging (EE5179)		- Modern computer vision (EE5178) - Vehicle dynamics (ED5220)	Introduction to field and service robotics (ED5315)
Programming Languages	C++, Python, C, Assembly	Libraries	Open3D, PCL, Eigen (C++), NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, OpenCV
Frameworks	PyTorch, ROS, MATLAB, Simulink	Miscellaneous	Git, Gerrit, Jenkins, Latex, Ubuntu
WORKSHOPS			
3D Vision – conducted by the Centre for Visual Information Technology, IIIT Hyderabad. Bayesian Estimation Methods in Robotics – delivered by faculty from University of Stuttgart at IIT Madras.			
MISCELLANEOUS	Teaching Assistant – Analog and Digital Electronics Laboratory, IIT Madras (July-Nov, 2025).		

1. Graduate Aptitude Test in Engineering, 2. Digital Signal Processors, 3. Monte Carlo-Extended Kalman Filter